Using Psychophysics: Thresholds and Signal Detection Theory:

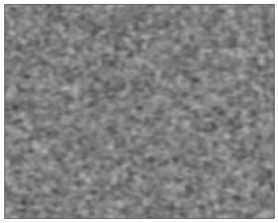
In this session you will learn how to set up, run and analyze a behavioral experiment in neuroscience. The experiment quantifies the relationship between physical stimuli and perception, a method commonly termed as psychophysics. Psychophysics can be combined with (and are often an essential part of) physiological measures such as fMRI, EEG etc.

We will run a short experiment to measure sensitivity for detecting weak stimuli masked in noise. We will test the minimum amount of contrast in a target that can still be reliably detected using two procedures to measure threshold: the staircase method and the method of constant stimuli. To test whether measures of threshold are a reliable measure of sensitivity, we will repeat the experiment three times with different levels of bias. Signal Detection Theory suggests that true sensitivity should not be affected by bias, but bias should affect the observer's response criterion. We will test this idea by calculating D' as a true measure of sensitivity.

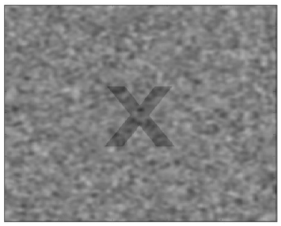
The experiment and the data analysis run in MATLAB using scripts I have provided. Only a small amount of input is required but try also to read the scrips to see what is happening. The target will be an X with noise added. Stimuli are presented for 0.05 seconds. The task is simple: If you see the target press X if you don't see it press spacebar. Check that CapsLock is off on your keyboard!

Target

Noise



Target + Noise



Include a description and discussion of the experiment, analysis and results in your practical report. Sit in your groups so that you can compare your results but prepare your report individually.

The steps of the experiment are:

1) Estimate your individual threshold using the staircase method.
2) Discuss in your group the design for the method of constant stimuli measures. Consider the number of stimulus conditions you will test and the order the three bias conditions
3) Collect data using the method of constant stimuli.
4) Analyze your individual data by calculating thresholds. Did your threshold change when you changed your bias? Does this mean you were more or less sensitive? Is that likely?
5) Analyze your individual data using Signal Detection Theory. Did sensitivity change or criterion?
6) Combine your group's data. What is the most stable measure of sensitivity, thresholds or D'?

For your report you must include:

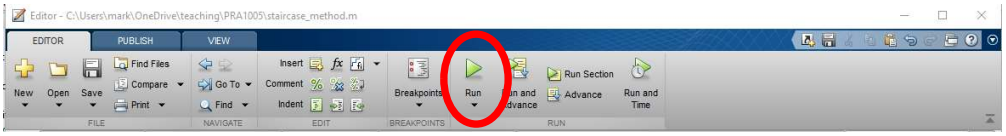
- at least 1 staircase figure.
- at least 1 individual constant stimulus figure.
- one figure comparing 3 psychometric functions.
- one figure and one table using signal detection theory.
- one figure that combines your group data.

All the figures you need are automatically made and saved during the session **but** made doubly sure you have what you need by pasting the figures into a document during the session.

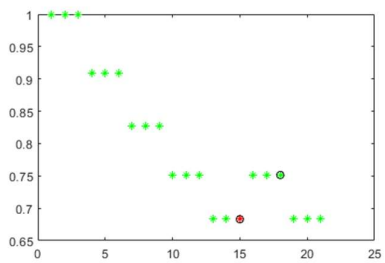
1) The Staircase Method

The first step to the experiment is get an estimate of how well you can perform the task. This can differ a lot between individuals and it depends on the settings of the screen and brightness of the room etc. The staircase method is an efficient method to estimate threshold. We will use this estimate to set the conditions used in the method of constant stimuli.

1. Open MATLAB and open the script '**staircase_method_2021.m**'. This script runs the experiment and will automatically save your data and some figures in the folder that contains the scripts.
2. Press Run



3. You might be asked to add the directory to the path. Click *add to path*
4. You will see a warning that the experiment will start and a countdown
5. You will see a grey screen followed by the stimulus (target or target + noise) and then a response screen. Press X or spacebar to register your response and start the next trial.
6. On a second figure you will see your progress through the staircase.



7. The experiment carries on until you make 20 **reversals** (not 20 trials).

When you make three correct judgments in a row the step increase (reduction in stimulus strength). If you make one error the step decreases (what percentage correct threshold will this give?). The step size gives the factor that the target contrast is divided by, so the signal-to-noise ratio is 1/step size. The threshold is calculated as the arithmetic mean of the last 14 reversal points

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threshold=(exp(mean(log(reversal(5:end)))))
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8. The staircase you just ran started above threshold and worked down. Would you get the same threshold if the staircase started below threshold and worked up? You could try by setting the start level to a high number. What start level would be good to choose? The first staircase could tell you.
9. If you want to repeat the measurement using the same start level. In that case tell MATLAB what repetition number you are running. Keep notes about what you did in each repetition.

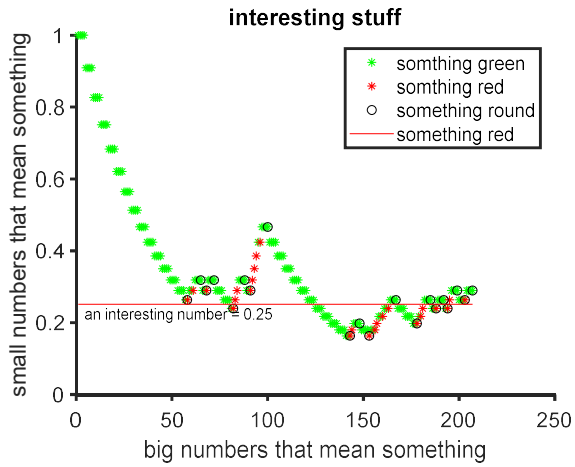
Analysis

The data (and figure) is saved into the data directory with a filename that includes the start level and the repetition number. Something like

I:/PRA1005/practical_1/staircase_start_level1_repetition1.mat

At the top of the script it checks whether the file is already written. If it does not exist, the experiment runs and the data is written. Otherwise the data is loaded and analyzed.

Go to the end of the script to see how to make and annotate the staircase figure correctly. You might need to move the legend box which you can do by clicking on it. Check that this figure is saved to your data folder. Include at least one staircase figure in your report to show how you decided your staircase threshold.



2) Setup for method of constant stimuli (group discussion)

In the staircase method, many stimulus levels are tested but each level is tested only a few times. The levels are chosen automatically by the staircase. In the method of constant stimuli, only a few levels are tested but they are tested many times. You must decide before starting an experiment which levels (called conditions in method of constant stimuli) will be tested and how many repetitions (trials) per condition you want to test.

If you choose many conditions and many trials per condition, the whole experiment will take a long time. If you choose few conditions and few trials per condition, the data will not be very reliable. We will fit a function to the data with **3 free parameters**. You must have more conditions that free parameters to avoid 'over fitting'. For every condition and repetition you add to your experiment you must count on adding two trials, one with and one without the target. So:

Total number of trials = Number of conditions X number for trials per conditions X 2

You can use the median time taken per trial during the staircase experiment to help you estimate how long the total experiment will take.

Estimated time in minutes = (Total number of trials X time per trial)/60

Consider that you will **repeat this measurement three times** for the three bias conditions**.** Aim to spend about half an hour collecting data.

When you have decided how many conditions you will test, you must also choose *which* conditions to test. You want some conditions for which you will have near-perfect performance, conditions with chance-level performance and some conditions in-between. You can use the threshold estimate from the staircase measurement to help: A stimulus that is twice your threshold will be easily visible while a stimulus that is half your threshold will be very difficult to see.

Plan your experiment together with the rest of your group. You should all use the same settings so that you can compare your results (number of conditions and number of trials per condition, each person should use their own threshold).

The measurement will be repeated three time under three levels of bias: 1) with neutral bias, 2) with bias to avoid misses and 3) with bias to avoid false alarms. Everyone in the group should participate in all three conditions (Unless you/they want to withdraw from the experiment, which is allowed for any reason!). As a group decide whether each person in the group should use the same order or should you balance them? Make sure to explain what you did in your report.

We will compare results from the group. Give each member an anonymous identification number (e.g. numbers between 1 and 6) that will be used to identify their data files.

3) Run the experiment

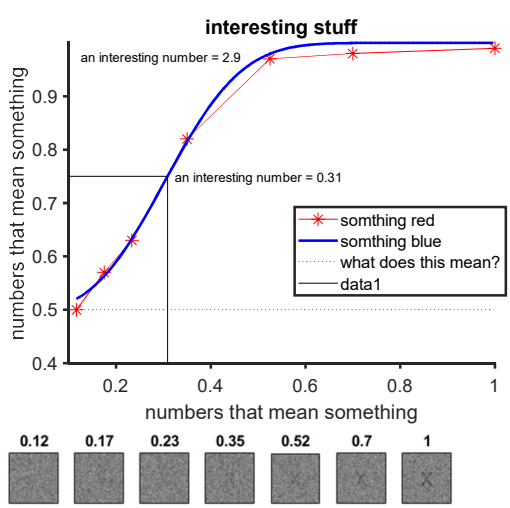
Once you are ready to start:

1. Open MATLAB script 'Constant stimuli_2021.m'
2. Set the data directory, your identification number and your estimated threshold.
3. Set what conditions you will test in terms of multiples of your threshold.
4. Set the number of trials per condition you will test.
5. Set the bias condition you will test
    1) In the *neutral* condition you get feedback if you make an error. Be as accurate as you can.
    2) In the *don't miss* condition you get feedback when you miss a target. Do your best but it you are not sure you should respond 'X'.
    3) In the *don't cry wolf* condition you get feedback when you report an X which was not there. Do your best, but don't report an X unless you are sure!
6. If you are repeating the measurement set the repetition number. Keep a note about what each you did in each repetition.
7. Press run to run the experiment.
8. Repeat the measurement with each bias condition in the order you decided.

4) Individual data analysis: Thresholds

Once you have completed all bias conditions, look at your data using the prepared script individual_thresholds_2021.m

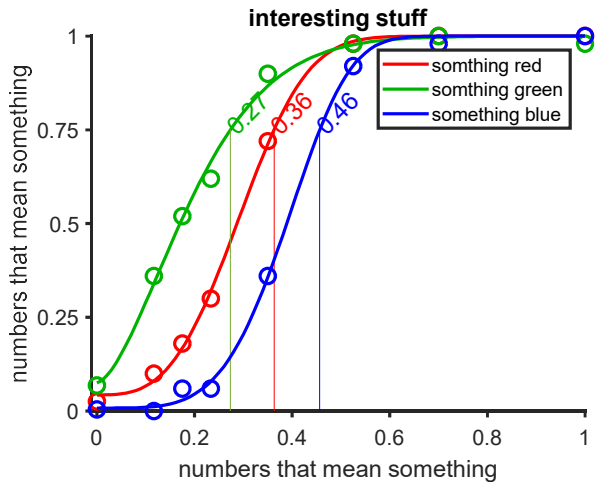
1. First look at the data from the neutral condition. Set the data directory, ID and repetition number that you want to load.



(Your figure may look different, I tend to change the colors and symbols year by year)

We fit a curve to the data to interpolate between the points that we measured. A standard function for a two alternative forced choice (called 2AFC) experiment is called the 'Weibull' cumulative distribution function. We use fminsearch to get the best fit of the function to the data. Think about the titles and what the numbers mean.

2. Next compare the three experiments to get a plot for the three risk conditions.



Make a table of the interesting numbers from the fitting for your report. What can you conclude the differences between the conditions?
Include the comparison figure with correct annotation and table in your report. Check that it is saved in your folder.

5) Individual data analysis: Signal detection theory

Open compare_experiments_SDT_2021.m

1. Did sensitivity really change between the experiments? Make a 2x2 table (in excel or word) from one condition showing the hit rate, miss rate, FA rate and correct rejection rate. The numbers appear in the MATLAB command window. Set the condition you want to show in the script as the display_CND

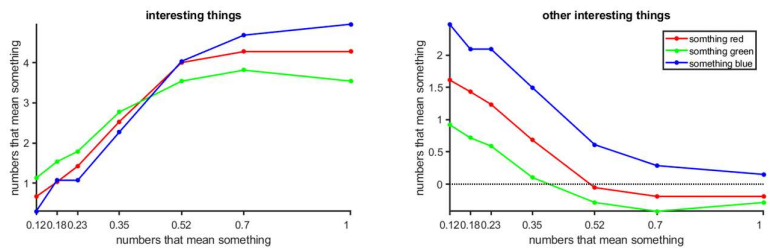
	Stimulus +	Stimulus -	Mean
YES	0.62	0.1	0.315
NO	0.38	0.9	0.64
	1	1	

The calculation of D' and Criterion are easy to do in MATLAB. Use MATLAB's norminv function to get a Z score probability for the hits and FA to calculate D' and criterion.

Hit rate=0.62, Z(Hit rate) = 0.3055
FA rate=0.1, Z(FA rate) =-1.2816
D' = Z(Hit rate)- Z(FA rate) = 1.587
Criterion= -(Z(Hit rate)+ Z(FA rate))/2 =-0.488

Include the table and calculation in your report

2. Finish making and annotating the figure

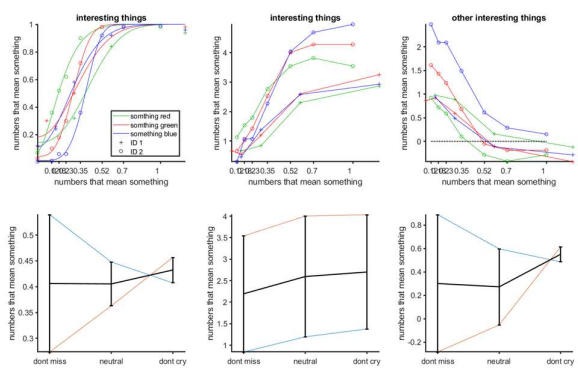


Include this figure in your report

3) Group analysis

Thresholds tend to be influenced by bias but measures of D' should be a true measure of sensitivity. Compare these measures from your group. Open Group_analysis.m

1. Put all the data from your group together in a folder.
2. Set the Data_directory and fill in a list of Participant_IDs and the Repetitions numbers you want to use for the three bias conditions. See further down the script how these numbers are used to set the correct filename for each datafile you want to use.
3. Set which stimulus level you want to display D' and Criterion from. Choose a mid-level condition close to the threshold.
4. Run the script and annotate the figure.



Include this figure in your report.

End of the practical

- In this practical you have worked with prepared scripts. If you are interested in improving your programming skills, you can read the script to understand each step.
- If you are interested in seeing more demonstrations of signal detection theory, you can explore ROC_demo.m.
- We used MATLAB with no special toolbox, to run psychophysics experiment. That is OK for our purposes but for true research you will want more control over accurate timing, stimulus size and luminance etc. For a controlled experiment you can use MATLAB Psychtoolbox http://psychtoolbox.org/

You can start working on your practical report already!